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Cover Letter: Student (m/f/d): Netlist Carpentry – Extending an interactive Circuit Viewer (IMMS_STUD_ME_1226)

IMMS GmbH Erfurt 2026-10-31

Dear Mr Schäfer,

I am applying as a student research assistant for the Netlist Carpentry topic. Extending an open-source Python framework with search, signal-path tracing and highlighting in a graph viewer is a concrete, well-scoped task, and it combines two things I have done before: working with graphs in Python and shipping tested, documented code.

I am an M.Sc. Data Science student at TU Ilmenau with a B.S. in Artificial Intelligence. I live and study in Ilmenau, so Erfurt is an easy commute.

- **Graphs in Python,** in a team project I extracted subject-relation-object triples from text into a NetworkX graph and compared graph and language-model embeddings
- **Tested, documented code,** my RAG project DocuMind has 37 offline unit tests, GitHub Actions CI, Docker and a README written for other users
- **Tracing problems,** I used an evaluation harness to trace both wrong answers in DocuMind to chunk boundaries, the same habit I would use for signal-path tracing

I would first learn the structure of Netlist Carpentry and the existing viewer, then add the search bar, signal-path tracing and node marking step by step, each with tests on example circuits and documentation that fits your existing suite. I have no microelectronics background, so I would rely on your circuits and team for the domain side and contribute the Python and graph work.

I can work up to 20 hours per week alongside my studies, starting immediately. I am also interested in your other Python topics, such as constant propagation for optimizing digital circuits. I would be glad to discuss the work.

I work in English (IELTS 8.0) and am currently learning German (A1).